

Prepared By	09-Jun-2012	Mr.Chatchawan S.	 RENEWCOILS ENGINEER & SUPPLY CO., LTD. Renew coils Engineering & Supply Co., Ltd.	
Checked By	09-Jun-2012	Mr.Chatchawan S.		
Approved By	09-Jun-2012	Mr.Amnauy W.		
Certified By	09-Jun-2012	Mr.Amnauy W.		
Rev.	Date	Checked	Approved	Description
0	09-Jun-2012	Mr.Chatchawan S.	Mr.Amnauy W.	Issue for Approved

Renew Coils Engineering & Supply Co., Ltd.

Doc. No. : RNC-WPS-002-2012

Addendum

**WELDING PROCEDURE SPECIFICATION (WPS)
RNC-WPS-005-1**

WELDING PROCEDURE SPECIFICATION (WPS)

Company Name **Renew Coil Engineering & Supply Co., Ltd.**

WPS No. **RNC-WPS-005-1**

Date **9-Jun-12**

Revision No. **0**

Supporting PQR No.(s) **RNC-PQR-005-1**

Date **9-Jun-12**

Type(s) **Manual**

Welding Process(es) **Gas Tungsten Arc Welding (GTAW)**

(Automatic, Manual, Machine, or Semi-Auto.)

JOINT (QW-402)

Joint Design **Single V**

Backing (Yes) ☐ (No) ☒ X

Backing Material (Type) **N/A**

☐ Metal ☐ Nonfusing Metal
☐ Nonmetallic ☐ Other

Notes _____

See Details Attachment sheet Joint Sketch

BASE METALS (QW-403)

P-No. **1** Group No. **1** to P-No. **1** Group No. **1**

Material Specification Type and Grade **SA 333 Gr.1.6 SA420 WPL 6** TO **SA 333 Gr.1.6 SA420 WPL 6**

Chem. Analysis and Mech. Prop. _____

Thickness Range :

Base Metal : Groove **1.6 mm - 7.82 mm.** Fillet **All**

Pipe Dia. Range : Groove **All** Fillet **All**

Other _____

FILLER METALS (QW-404)

Spec. No. (SFA) **5.28**

AWS No. (Class) **ER-70S-G**

F-No. **6**

A-No. **1**

Size of Filler Metals **Ø 2.0 & 2.4 mm**

Deposited Weld Metals **3.91 mm**

Thickness Range :

Groove **7.82 mm Max.**

Fillet **All**

Electrode-Flux (Class) **-**

Flux Trade Name **-**

Consumable Insert **-**

Other **KobeTGS-1N or Equivalents**

POSITION (QW-405)

Position(s) of Groove **ALL**

Welding Progression Up ☒ X Down ☐

Position(s) of Fillet **ALL**

POSTWELD HEAT TREATMENT (QW-407)

Temperature Range **N/A**

Time Range **N/A**

Notes _____

PREHEAT (QW-406)

Preheat Temp. Min. **10°C**

Interpass Temp. Max. **238°C**

Preheat Maintenance **N/A**

Notes _____

GAS (QW-408)

Percent Composition

	Gas(es)	(Mixture)	Flow Rate
Shielding	Argon	99.99%	13-17Ltr/Min
Trailing	-	-	-
Backing	-	-	-

WELDING PROCEDURE SPECIFICATION (WPS)

WPS No. RNC-WPS-005-1 Rev. 0

ELECTRICAL CHARACTERISTICS (QW-409)

Current ☐ AC ☒ DC Polarity EN
 Amps (Range) 80-120 Volts (Range) 10-13
 Tungsten Electrode Size and Type Ø 2.4 mm, Thoriated (EWTh-2)
 (Pure Tungsten, 2% Thoriated, etc.)
 Mode of Metal Transfer for GMAW N/A
 (Spray arc, short circuiting arc, etc.)
 Electrode Wire feed speed range N/A

TECHNIQUE (QW-410)

String or Weave Bead BOTH APPLICATIONS
 Orifice or Gas Cup Size 8.0 mm - 19.0 mm
 Initial and Interpass Cleaning (Brushing, Grinding, etc.) BRUSHING OR GRINDING
 Method of Back Gouging N/A
 Oscillation N/A
 Contact Tube to Work Distance N/A
 Multiple or Single Pass (per side) MULTIPASS
 Multiple or Single Electrodes SINGLE ELECTRODE
 Travel Speed (Range) 5.0-8.0Cm/Minute
 Peening N/A
 Other N/A

Weld Layer(s)	Process	Filler Metal		Current		Volt Range	Travel Speed Range (Cm/min)	Other
		Class	Dia. (mm)	Type Polar.	Amp. Range			
1st	GTAW	ER70S-G	2.4	DC EN	80-100	10-12	5.0-7.0	
2nd & Other(s)	GTAW	ER70S-G	2.4	DC EN	90-120	11-13	6.0-8.0	

We, the undersigned, certify that the statements in this record are correct and that the test welds were prepared, welded, and tested in accordance with the requirement of **ASME IX, 2007 latest Revision**.

Prepared by:

Mr. Chatchawan S.
Name

Signed

9/06/12
Date

Reviewed by:

Mr. Sansan ch
Name

Signed

9-06-12
Date

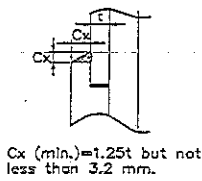
Approved by:

ANNVAY W.
Name

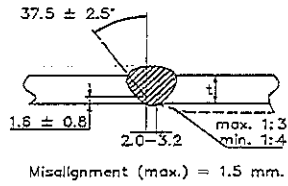
Signed

29-06-12
Date

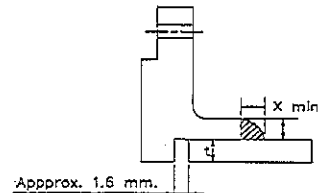
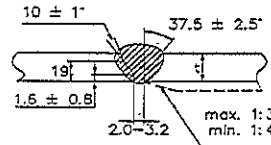
ATTACHED FIGURE JOINT SKETCH



JOINT No. A1

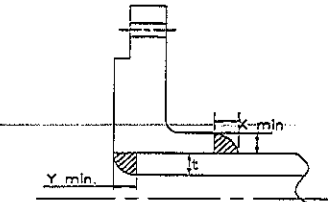


JOINT No. B1 & B2

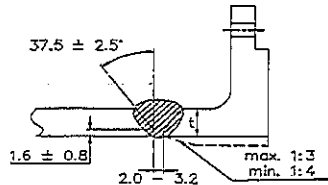


JOINT No. A2

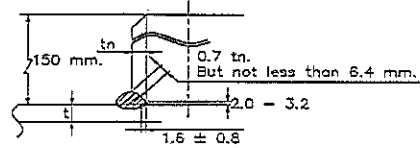
X min. = The lesser of 1.4t or thickness of the hub.



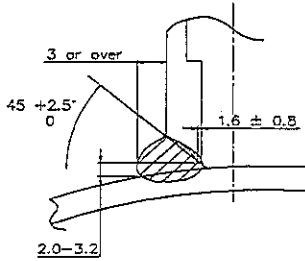
JOINT No. A3



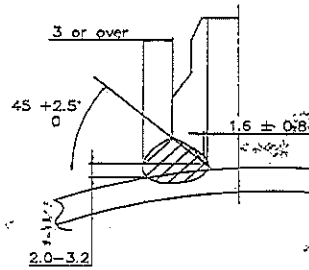
JOINT No. B3



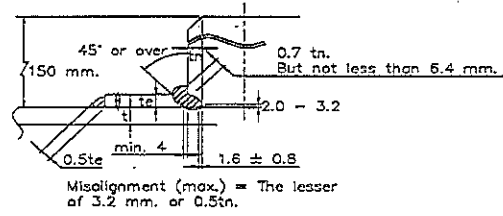
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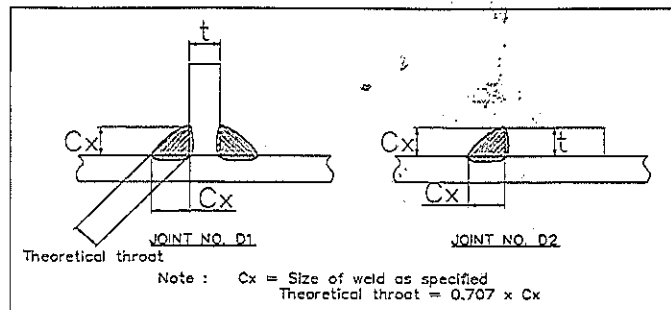
JOINT No. C1



JOINT No. C2



JOINT No. B5





Procedure Qualification Record (PQR)

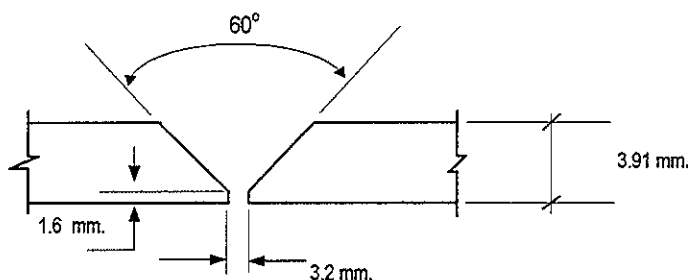
ASME-QW-200.2, Section IX, ASME Boiler Pressure Vessel Code

Record Actual Variables Used to Weld Test Coupon

RECORD OF WELDING (QW-483) PQR Number : PQR-RNC-005-1

Company Name :	RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.		
PQR Record Number :	RNC-PQR-005-1	Date :	13 June 2012
WPS Number :	RNC-WPS-005-1		
Welding Process (es) :	Gas Tungsten Arc Welding (GTAW)		
Type (Manual, Semi-auto, or Automatic) :	Manual		

JOINT DESIGN (QW-402)



(For combination qualifications, the deposited weld metal thickness shall be recorded for each filler metal or process used)

WELDING VARIABLES

BASE METALS (QW-403)				POSTWELD HEAT TREATMENT (QW-407)																			
Material Spec.	SA333	To	SA333	Temperature	N/A																		
Type or Grade	Gr.6	To	Gr.6	Hold Time	N/A																		
P. No.	1	To	1	Others	GAS (QW-408) Percent Composition <table><tr><td></td><td>Gas (es)</td><td>Mixture</td><td>Flow Rate</td></tr><tr><td>Shielding</td><td>Argon</td><td>99.95%</td><td>15 L/Min.</td></tr><tr><td>Trailing</td><td>-</td><td>-</td><td>-</td></tr><tr><td>Backing</td><td>-</td><td>-</td><td>-</td></tr></table>				Gas (es)	Mixture	Flow Rate	Shielding	Argon	99.95%	15 L/Min.	Trailing	-	-	-	Backing	-	-	-
	Gas (es)	Mixture	Flow Rate																				
Shielding	Argon	99.95%	15 L/Min.																				
Trailing	-	-	-																				
Backing	-	-	-																				
Thickness of Test Coupon	3.91 mm.																						
Diameter of Test Coupon	Ø 2"																						
Others																							
FILLER METALS (QW-404)				ELECTRICAL CHARACTERISTICS (QW-409)																			
SFA Specification	A 5.28			Current	DC																		
AWS Classification	ER70S-G			Polarity	EN																		
Filler Metal F-No.	6			Amperage	85 - 94	Voltage	10 - 11																
Weld Metal Analysis A-No.	1			Tungsten Electrode Size	ø 2.4 mm. 2% thoriated.EWTh-2																		
Size of Filler Metal	Ø 2"			Others	N/A																		
Others	Kobelco TGS-1N			TECHNIQUE (QW-410)																			
Weld Metal Thickness	-			Travel Speed	5 - 7 cm/min.																		
POSITION (QW-405)				String or Weave Bead	Both																		
Position of Groove	6G			Oscillation	N/A																		
Progression of Welding (Uphill or Down)	Up Hill			Multipass or single Pass	Multiple Pass																		
Others	N/A			Multipass or Single Electrodes	Single Electrode																		
PREHEAT (QW-406)				Others	N/A																		
Preheat Temperature	Ambient Temperature																						
Interpass Temperature (max.)	220 °C																						
Others	N/A																						

Procedure Qualification Record

ASME-QW-200.2, Section IX (WPQR)

RECORD OF TEST RESULTS (QW-483) PQR Number:- RNC-PQR-005-1

Tensile Test Report No.TS-12-06-103

Specimen	Width (mm)	Thickness (mm)	Area (mm ²)	Ultimate Total Load (KN)	Ultimate Strength (N/mm ²)	Type of Failure & Loc.
2364/12 TR1	13.10	3.49	45.719	22.163	484.766	OUT OF WELD
2364/12 TR2	13.08	3.08	48.134	22.202	461.255	OUT OF WELD

Guided Bend Tests (QW-160) Report No.BD-12-06-101

Type and Figure Number	Type of Bend	Indication	Results
2364/12 BF1	Transverse face bend	No Open Discontinuity	Accepted
2364/12 BF2	Transverse face bend	No Open Discontinuity	Accepted
2364/12 BR1	Transverse root bend	No Open Discontinuity	Accepted
2364/12 BR2	Transverse root bend	No Open Discontinuity	Accepted

Toughness Tests (QW-170)

Specimen Number	Notch Location	Notch Type	Test Temp.	Impact Values	Lateral Expansion		Drop Weight	
					% Shear	Mils	Break	No Break
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Fillet Weld Test (QW-180)

Result - acceptable : Yes N/A No N/A Penetration Into Base Metal : Yes N/A No N/A
Macroetch Test Results N/A

Other Tests

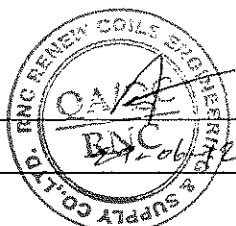
Type of Test Radiographic examination results Accepted Report No.PQR -002
Hardness Testing Report No. HN-12-06-078
Macrostructure Report No. MA-12-06-100
Impact Testing Report No.IM-12-06-137

Welders Name Mr.Decha T. Welder Number -
Tests Conducted By Mr.Keerati Sa-ngoen Laboratory Test LPN PLATE MILL

We certify that the statements in this record are correct and that the test coupons were prepared, welded, and tested in accordance with the requirements of section IX of the ASME Boiler and Pressure Vessel Code.

Manufacture or Contractor :

Date :



Inspected by : Mr.Natthawut M.

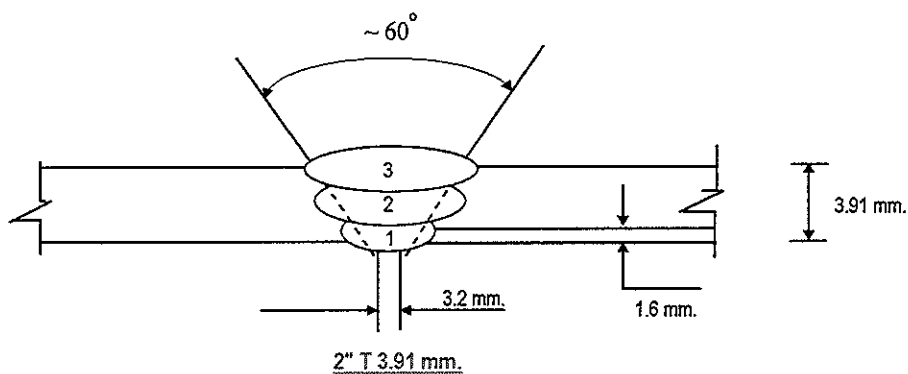
Certified by : Mr.Chalermkiet R.

Date :



PAE TECHNICAL SERVICE	WELDING PROCEDURE	PQR No. : RNC-PQR-005-1
		WPS No. : RNC-WPS-005-1
	QUALIFICATION RECORDS	THIS PQR RECORDS ACTUAL TEST
		CONDITION AND RESULT

Welding Sequences for PQR-RNC-005-1 (Butt Joint)



Weld Bead No.	Process	Class Filter Metal	Wire Size (mm)	Arc Current (Amps)	Arc Voltage (Volts)	Pot. AC/DC+	String or	Wire Feed	Rate of Travel (cm/min)	Heat Input (KJ/cm)	Gas Shielding Flow Rate (l/min)	Gas Backing Flow Rate (l/min)
							Weave Bead	Speed				
1	GTAW	ER70S-G	2.4	85	10	DCEN	-	-	5	10.20	15	-
2	GTAW	ER70S-G	2.4	91	11	DCEN	-	-	6	10.01	15	-
3	GTAW	ER70S-G	2.4	94	11	DCEN	-	-	7	8.86	15	-



Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 - 9978

Radiographic Examination Report

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Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 - 9978

Mechanical Test Report



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmp.co.th



21711

Tension Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.
Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Page 1 of 1
Report No. TS-12-06-103
Revision No. 0

Test Product	PQR No. RNC-PQR-005-1	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 (Prepared by ASME IX : 2010)
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010
Material Specification	A333 Gr.6	Type of Test	Reduced Section Tension
Dimension	Ø 2" Sch 40 (3.91 mm Thk.)	Equipment's Capacity	200 TONS
Date Received	19 June 2012	Equipment / Serial No.	SHIMADZU UH200A / 32098926
Date Tested	21 June 2012	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Specimen Dimension					Yield		Ultimate		Elongation (%)	Remark
		Diameter (mm)	Thickness (mm)	Width (mm)	Area (mm ²)	Gauge Length (mm)	Load (kN)	Strength (N/mm ²)	Load (kN)	Strength (N/mm ²)		
1	2364/12 TR1	N/A	3.49	13.10	45.719	N/A	N/A	N/A	22.163	484.766	N/A	OUT OF WELD
2	2364/12 TR2	N/A	3.68	13.08	48.134	N/A	N/A	N/A	22.202	461.255	N/A	OUT OF WELD

The unit of N/mm² is equivalent to MPa

Additional details : WPS No. RNC-WPS-005-1

Welding Process : GTAW

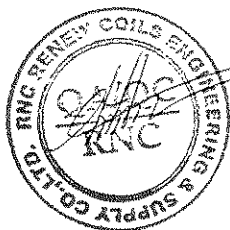
Filler Metal/Electrode Spec't : ER-70S-G/TGS-1N

Reported by :

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date 29 / 06 / 12



Approved by :

(Mr.Pavaret Freedawiphat)

Deputy Managing Director

Date 29 / 06 / 12

LPN PLATE MILL

Notes 1. The above results are valid exclusively for tested samples as mentioned in this report.

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LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmpm.co.th

21719

Bend Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA AMUANG, RAYONG 21000

Report No. BD-12-06-101

Revision No. 0

Test Product	PQR No. RNC-PQR-005-1	Material Characterization	Pipe
Project Name	N/A	Test Method	ASME Sec. IX : 2010
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010
Material Specification	A333 Gr.6	Type of Jig	Guided-Bend Roller Jig
Dimension	Ø 2" Sch 40 (3.91 mm Thk.)	Diameter of Mandrel	16 mm
Date Received	19 June 2012	Bend Angle	180 Degree.
Date Tested	21 June 2012	Equipment / Serial No.	SHIMADZU UH200A/32098926

Item No.	Test No.	Type of Bend	Size of Specimen (ThickxWidthxLength)	Indication	Location of Indication	Remark
1	2364/12 BF1	Transverse face bend	3.89x19.00x160.13 mm	No Open Discontinuity	N/A	-
2	2364/12 BF2	Transverse face bend	3.82x19.00x163.39 mm	No Open Discontinuity	N/A	-
3	2364/12 BR1	Transverse root bend	3.85x19.00x162.55 mm	No Open Discontinuity	N/A	-
4	2364/12 BR2	Transverse root bend	3.87x19.00x160.17 mm	No Open Discontinuity	N/A	-

Additional details : WPS No. RNC-WPS-005-1

Welding Process : GTAW

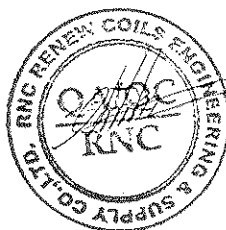
Filler Metal/Electrode Spec't : ER-70S-G/TGS-1N

Reported by : N. J. P. D. S.

(Mr. Keerati Sa-ngoen)

Laboratory Section Manager

Date 29/06/12



Approved by : N. J. P. D. S.

(Mr. Pavaret Preechawiphat)

Deputy Managing Director

Date 29/06/12

LPN PLATE MILL

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Testing Laboratory LPN Plate Mill Public Company Limited

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Tel. 0 2815 6400-9 ext. 1179,1210, Direct Line and Fax. 0 2815 6415, E-mail : qa@lpnmp.co.th



2865

Impact Testing Report

Customer : RENEWCOILS ENGINEERING & SUPPLY CO., LTD.

Page 1 of 1

Address : 59 MOO 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000

Report No. IM-12-06-137

Revision No. 0

Test Product	PQR No. RNC-PQR-005-1	Material Characterization	Pipe
Project Name	N/A	Test Method	ASTM A370-10 and ASTM E23M-07a ^{nl}
Project No.	N/A	Standard of Test	ASME Sec. IX : 2010
Material Specification	A333 Gr.6	Type of Test	Charpy V-Notch Impact
Dimension	Ø 2" Sch 40 (3.91 mm Thk.)	Equipment / Serial No.	Tinius Olsen / 221811
Date Received	19 June 2012	Equipment 's Capacity	542 Joules
Date Tested	22 June 2012	Ambient Temperature	23 ± 5 °C

Item No.	Test No.	Test Temperature (°C)	Size of Specimen			Results		Location of Test	Remark
			Thickness (mm)	Width (mm)	Length (mm)	Energy Absorbed (Joules)	Average (Joules)		
1	2364/12-1	-20°C	2.524	10.012	54.79	29.84	29.54	Base Metal	-
2	2364/12-2	-20°C	2.527	10.012	54.77	30.86			
3	2364/12-3	-20°C	2.524	10.011	54.75	27.92			
4	2364/12-1	-20°C	2.507	10.009	54.89	41.67	40.21	Weld Metal	-
5	2364/12-2	-20°C	2.503	10.006	54.89	40.52			
6	2364/12-3	-20°C	2.513	10.004	54.99	38.44			
7	2364/12-1	-20°C	2.508	10.011	54.78	42.62	43.88	HAZ	-
8	2364/12-2	-20°C	2.507	10.007	54.77	46.41			
9	2364/12-3	-20°C	2.505	10.008	54.79	42.62			

Additional details : WPS No. RNC-WPS-005-1

Welding Process : GTAW

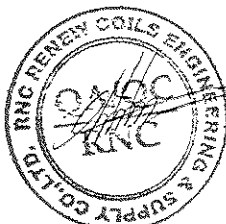
Filler Metal/Electrode Spec't : ER-70S-G/TGS-1N

Reported by :

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date 30 / 07 / 12

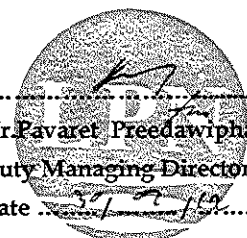


Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date 31 / 07 / 12

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LPN PLATE MILL



LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

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11063

Macrostructure Analysis Report

Page 1 of 1

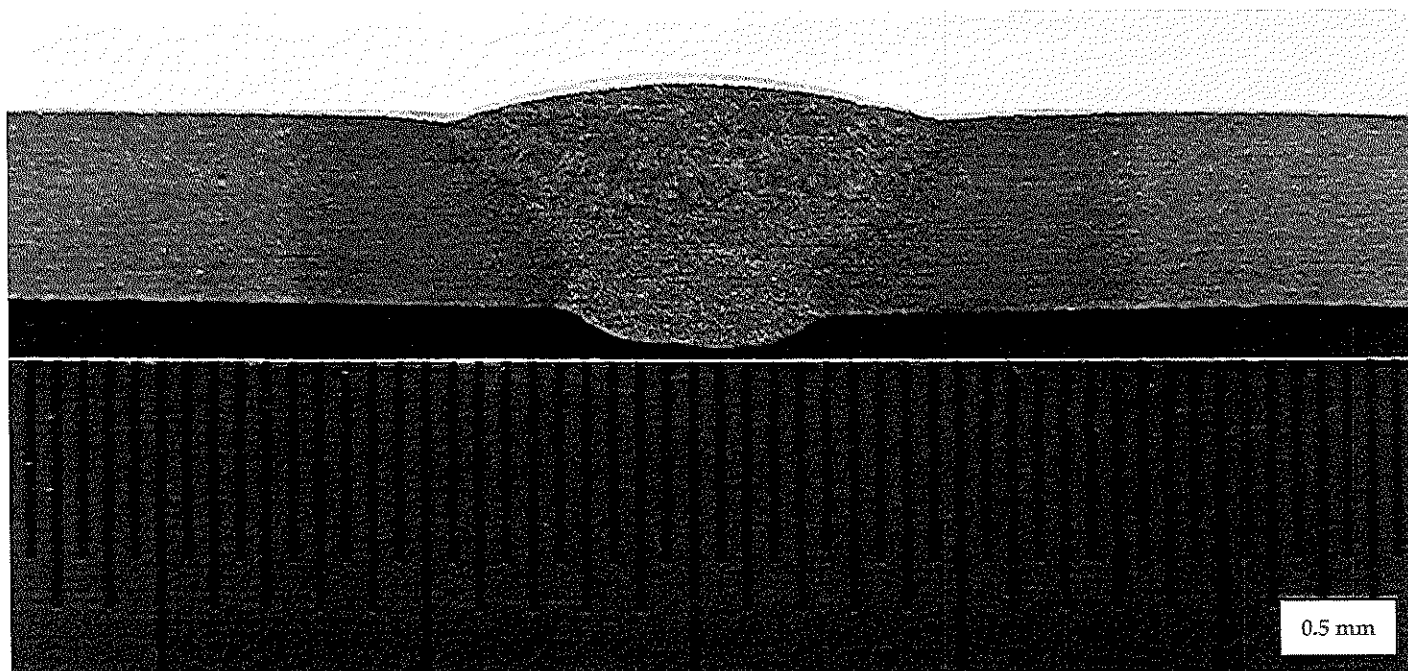
Customer : RENEWCOILS ENGINEERING & SUPPLY CO.,LTD.

Report No. MA-12-06-100

Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG, RAYONG 21000 Thailand.

Revision No. 0

Test Product	PQR No. RNC-PQR-005-1	Material Characterization	Pipe
Test No.	2364/12 MA	Test Method	ASME IX : 2010
Project Name	N/A	Standard of Test	ASME IX : 2010
Project No.	N/A	Etchant Reagent	75ml H ₂ O+25ml HNO ₃ (65%)
Material Specification	A333Gr.6	Temperature	28 °C
Dimension	Ø2"Sch40(3.91 mm Thk.)	Time of Etching	5 minutes
Welding Process/Position	GTAW	Surface Preparation	Polishing with sand paper
Type of Joint	Butt Joint		240, 400, 600, 800, 1000
Date Received	19 June 2012	Date Tested	21 June 2012



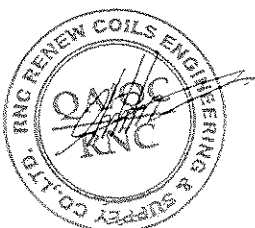
Additional details : The macrostructure shows the complete fusion.Weld metal and heat affected zone were free of cracks
Filler/Electrode Spec't : ER-70S-G TGS-1N

Reported by :

(Mr.Keerati Sa-ngoen)

Laboratory Section Manager

Date : 29/06/12



Approved by :

(Mr.Pavaret Preedawiphat)

Deputy Managing Director

Date : 29/06/12

LPN PLATE MILL

Notes

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LPN PLATE MILL

Testing Laboratory LPN Plate Mill Public Company Limited

12405

199/9 Moo 4, Suksawad Rd., Pakklongbangplakod, Prasamutjedee, Samutprakarn 10290 Thailand.

Tel. 0 2815 6400-9 ext. 1179, 1210, Direct Line Fax. 0 2815 6415 , E-mail : qa@lpnpm.co.th

Hardness Testing Report

Page 1 of 1

Customer : RENEWCOILS ENGINEERING & SUPPLY CO.,LTD..

Report No. HN-12-06-078

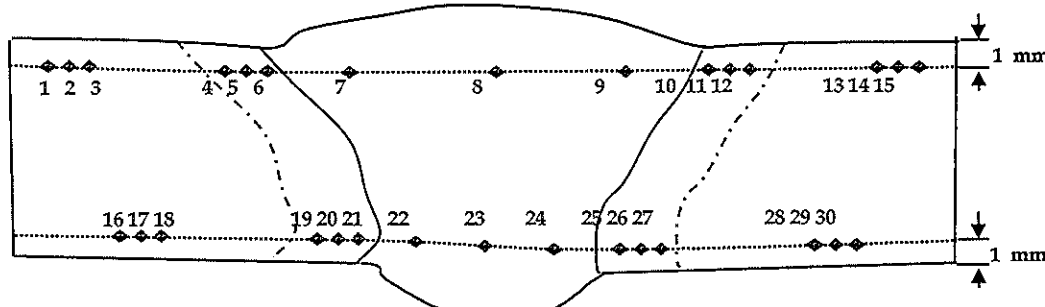
Address : 59 Moo 8 HIGHWAY 36 T.TABMA A.MUANG,RAYONG 21000 Thailand.

Revision No. 0

Test Product	PQR No. RNC-PQR-005-1	Material Characterization	Pipe
Test No.	2364/12 HN	Test Method	ASTM E92-82 : 2003
Project Name	N/A	Standard of Test	ASTM Sec. IX : 2010
Project No.	N/A	Type of test	Vicker Hardness Test
Material Specification	A333Gr.6	Equipment	Mitutoyo, HV-115
Dimension	Ø2"Sch40(3.91 mm Thk.)	Indenter Size	Diamond, pyramid 136 deg.
Welding Process/Position	GTAW	Test Load	HV 10 Kgf.
Type of Joint	Butt Joint	Total Force Dwell Time	15 s
Date Received	19 June 2012	Cal. Block	No. 62137 No. A62072
Date Tested	21 June 2012	Ambient Temperature	25.8 °C

Test Location

Total = 30 points



Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale HV)	Point No.	Location	Hardness value Value (Scale)
1	BASE	132.0	16	BASE	137.9	*****		
2	BASE	131.0	17	BASE	137.6			
3	BASE	131.2	18	BASE	137.1			
4	HAZ	146.4	19	HAZ	151.1			
5	HAZ	147.3	20	HAZ	160.5			
6	HAZ	150.9	21	HAZ	170.5			
7	WELD	195.2	22	WELD	156.6			
8	WELD	199.5	23	WELD	158.0			
9	WELD	189.6	24	WELD	156.3			
10	HAZ	166.4	25	HAZ	169.6			
11	HAZ	157.6	26	HAZ	163.6			
12	HAZ	146.8	27	HAZ	157.5			
13	BASE	131.7	28	BASE	141.6			
14	BASE	132.4	29	BASE	140.5			
15	BASE	131.4	30	BASE	142.0			

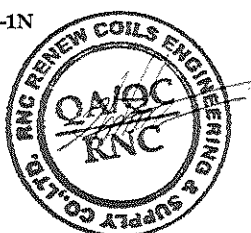
Additional details : Filler/Electrode Spec't : ER-70S-G TGS-1N

Reported by :

(Mr. Keerati Sa-ngoen)

Laboratory Section Manager

Date : 29/06/12

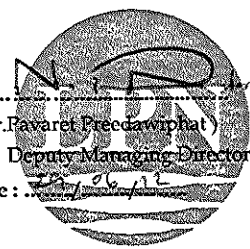


Approved by :

(Mr. Pavaret Preedawiphat)

Deputy Managing Director

Date : 29/06/12



LPN PLATE MILL

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Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 - 9978

Consumable Certificate

TKW110801 4520000

INSPECTION CERTIFICATE
(検査証明書)PURCHASER
需 要 家WELDING ROD OR WIRE
溶加棒又はワイヤCERTIFICATE No.
証明書番号: 5201132-0
DATE OF ISSUE
発行日: 2011.10.28

TRADE DESIGNATION 品 名 (型 番)	DIMENSION 寸 法 (mm)	MFG.No 製造番号	APPLICABLE SPECIFICATION AND CLASSIFICATION 適用規格及び種類
TG-S1N	(2.4) mm 3/32 inch	GZ00820	AWS A5.28 ER70S-G ASME SFA-5.28 ER70S-G

CHEMICAL COMPOSITION 化学成分(%) ACCORDING TO EN10204 TYPE 3.1

ELEMENTS 成分	C	SI	MN	P	S	CU	NI	MO					
ROD OR WIRE 棒又はワイヤ	0.06	0.31	1.04	0.006	0.009	0.11	0.80	0.14					
ELEMENTS 成分									FS = FERRITE % (SCHAEFFLER DIAGRAM)				
ROD OR WIRE 棒又はワイヤ									FN = FERRITE NUMBER (DELONG DIAGRAM)				
									FNW = FERRITE NUMBER (WRC)				

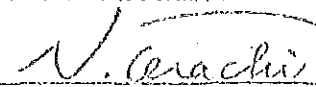
TENSILE PROPERTIES OF DEPOSITED METAL 溶着金属の引張性能				IMPACT PROPERTIES OF DEPOSITED METAL 溶着金属の衝撃性能			HARDNESS PROPERTIES 硬さ
YIELD POINT 降伏点	YIELD STRENGTH AT 0.2% OFFSET 0.2% 耐力	TENSILE STRENGTH 引張強さ	ELONGATION 伸び	TEST TEMP. 試験温度	ABSORBED ENERGY 吸収エネルギー		
(-) N/mm ² ksi	(≥360) N/mm ² ≥52 ksi	(≥480) N/mm ² ≥70 ksi	≥24 %	(-60) °C -76 °F	AVG. 平均 (≥27) J ≥20 ft-lbf	EACH (≥21) J ≥15	

WELDING CONDITIONS 溶接条件				POSTWELD HEAT TREATMENT 溶接後熱処理
TYPE OF CURRENT 電流の種類	DCEN	SHIELDING GAS シールドガス	AR 100%	
AMPERAGE 溶接電流	220 A			(-) °C X - h - °F X - h
ARC VOLTAGE アーク電圧	14 V			

WE HEREBY CERTIFY THAT THE ABOVE TEST RESULTS
AND MECHANICAL PROPERTIES SPECIFICATIONS SATISFY
THE REQUIREMENTS OF THE APPLICABLE SPECIFICATION.

上記試験結果および機械的性質保証値は、適用規格の要求を
満足していることを証明します。

◆KOBEL STEEL, LTD.
WELDING BUSINESS FUKUCHIYAMA PLANT
株式会社 神戸製鋼所 溶接事業部門 福知山工場



APPROVED BY CHIEF INSPECTOR
主任検査員 承認済



Renew Coils Engineering & Supply CO., LTD.

59 MOO 8 HIGHWAY 36, T. TABMA, A. MUANG, RAYONG 21000

TEL : (038) 663 - 476 FAX : (038) 633 - 477 MOBILE : (086) 340 - 9978

Material Certificate

INSPECTION CERTIFICATE EN10204-3.1

Issued by
BENTELER STAHL/ROHR GMBH
RESIDENZSTR. 1
D-33104 PADERBORN,
GERMANY

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BENTELER

Stahl/Rohr

ABNAHMEPRÜFZEUGNIS EN 10204-3.1

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CERTIFICAT DE RECEPTION EN 10204-3.1
EN 10204:2006-01

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Kunden-Bestell-Nr.: RJ811071RM
Purchase Order No.:
No. de commande du client:

Hersteller:
Manufacturer:
Producteur: BENTELER STAHL/ROHR GMBH

Herstellerzeichen:
Manufacturer's brand:
Marque du producteur:



Stempel des Abnahmebeauftragten: WA
Stamp of the inspection representative:
Poinçon du contrôleur:

Stahlschmelzungsverfahren: ELEKTROSTAHL
Steelmaking process: ELECTRIC FURNACE
Procédé d'élaboration de l'acier: FOUR ELECTRIQUE

ROJPAIBOON EQUIPMENT CO.,LTD.
303,305,307,309 SIRINTORN RD.,
BANGBUMRUH, BANGPLAD,
BANGKOK 10700, THAILAND

Order confirmation no. 1538048

Versandanzahlen-Nr.: 6528487
Dispatch Note No.:
No. d'avis d'expédition:

Seamless carbon steel pipe ASTM A333 grade 6 / ASME
SA333 NACE MR0175 details as per order confirmation
no 1538048 date 12.11.2008

L/C NUMBER: ML09000381

Lieferbedingungen: ASTM-A 333-2005-HF, ASME SA-333-HF, ASME Section II Part A Edition 2007 Addenda 2008, NACE
Terms of delivery:
Conditions de livraison: MR0175/ISO 15156-1:2001, NACE MR0175/ISO 15156-2:2003

Maße - Toleranzen: length acc. to customer request, ASME B36.10M-2004, ASME SA-999, ASME Section II Part A Edition 2007 Addenda
Dimensions-tolerances: 2008, ASTM-A 999-2004a
Dimensions-tolérances:

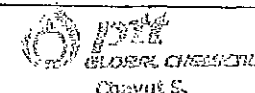
Stahlsorte: GRADE 6

Steel grade:
Nuance d'acier:

Lieferzustand: Normalized

Delivery condition:
État de livraison:

Produktkennzeichnung: FS: BENTELER SIGN BENTELER A333-HF-SA333 6 SMLS SCHEDULE LT 50 WA TUBES-LENGTH METERS
Product marking: DIMENSIONS HEAT-NO. NACE MR 0175/ISO 15156-1/2 - RM0811
Marquage du produit:



Inspection QA/QC Division
Integrity and Reliability

Chuyut S.

AEZ = Ätzzeichenbeschriftung, Etching ink marking, Gravure à l'encre FK = Fechtzeichenzeichnung, colour marking, marquage par couleur FS = Farbschablonierung, paint stencilling, marquage par peinture FSD = Farbschablonierung, Colour jet printing, imprimante à jet d'encre de c
LS = Längsschablonierung, longitudinal stencilling, marquage longitudinal par peinture FLE = Etikettenkennzeichnung, tag marking, marquage sur étiquette PS = Prägestempel, die stamp, marquage par poinçonnage TS = Tixensschablonierung, ink jet spray marking, imprimante à jet d'encre

Pos.	Stück	Maße	Länge	Gewicht	Schmelzen-Nr.	Prüfdruck	Rohr-Nr.-Gruppe	Vielfachlängen
Item	Number	Dimensions	Length	Weight	Heat No.	Test pressure	Tube number group	Multiple lengths
Poste	Nombre	Dimensions	Longueur	Poids	No. de coulée	Pression d'épreuve	Série de no. des tubes	Longueurs multiples
		mm	m	kg				
0010	82	2" NPS * Sched. 40	6000	492,00	2664	552918		

PO 15019701

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Pos.	Stück	Maße	Länge	Gewicht	Schmelzen-Nr.	Prüfdruck	Rohr-Nr.-Gruppe	Vielfachlängen
Item	Number	Dimensions	Length	Weight	Heat No.	Test pressure	Tube number group	Multiple lengths
Poste	Nombre	Dimensions	Longueur	Poids	No. de coulée	Pression d'épreuve	Série de no. des tubes	Longueurs multiples
		mm	m	kg				
0020	103	3" NPS * Sched. 40	6000	618,00	7067	762106		
0030	134	4" NPS * Sched. 40	6000	904,00	13070	762135		

Schmelzenanalyse [%] / Heat analysis [%] / Analyse sur coulée [%]

Pos.	Schmelzen-Nr.	C	SI	MN	P	S	AL
Item	Heat No.						
Poste	No. de coulée						
0010	552918	0,130	0,210	0,83	0,010	0,001	0,032
0020	762106	0,130	0,215	0,78	0,011	0,002	0,032
0030	762135	0,125	0,220	0,77	0,009	0,003	0,030

 GLOBAL CHEMICAL
Chayut S.
Inspection CA/RC Division
Integrity and Reliability

Prüfergebnisse / Test results / Résultats des essais

Die Rohre wurden zerstörungsfrei geprüft:

The tubes are non destructively tested:

Les tubes ont passé un essai non destructif:

ET-test: acc. to ASME SA-999, ASTM-A 999; acc. to ASTM-E 309; imperfections

PASSED

Augensichtkontrolle:

PASSED

Maßkontrolle:

PASSED

Ringfaltversuch:

PASSED

Visual inspection:

Dimensions examination:

Flattening test:

Examen visuel:

Vérification des dimensions:

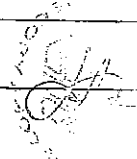
Essai d'aplatissement:

Ergebnisse der mechanischen Prüfung / Results of mechanical testing / Résultats des essais mécaniques

Die Probenahme erfolgte an Vielfachlängen.

The sampling was carried out on multiple lengths.

L'échantillonnage était réalisé aux longueurs multiples.



PC 15015363

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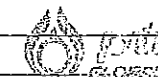
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Page:

Page:

Zugversuch Streifenprobe / Tensile test Strip test specimen / Essai de traction Bande decoupee sur tube

Pos. Item Poste	Proben-Nr. Specimen No. No. de l'éprouvette	Schmelzen-Nr. Heat No. No. de coulée	Probenabmessung Specimen dimensions Dimensions de l'éprouv.	Streckgrenze Yield strength Limite élastique	Zugfestigkeit Tensile strength Résistance à la traction	Dehnung Elongation Allongement	Einschnürung Area reduction Coefficient de striction	Re/Rm Re/Rm Re/Rm
Anforderungen Requirements Exigences			mm	ReH N/mm ² MIN 240	Rm N/mm ² 415-750	A2° % MIN 27	1. Formel 1. Formula 1 Formule	
0010	000001	552918	19,00 X 4,00	330	473	36		
0010	000002	552918	19,00 X 4,00	320	474	34		
0020	000001	762106	25,40 X 5,60	318	467	40		
0020	000002	762106	25,40 X 5,50	326	471	42		
0030	000001	762135	25,40 X 5,90	317	483	36		
0030	000002	762135	25,40 X 6,10	320	477	40		
0030	000003	762135	25,40 X 6,00	338	486	42		



Chayot S.

Inspection QA/QC

Integrity and Reliability

Kerbschlagbiegeversuch Notched bar impact test / Essai de flexion par choc (résilience) [CHARPY_V]

Pos. Item Poste	Proben-Nr. Specimen No. No. de l'éprouv.	Schmelzen-Nr. Heat No. No. de coulée	Probenabmessung Specimen dimensions Dimensions de l'éprouvette	Probenlage Specimen position Position de l'éprouvette	Prüftemperatur Test temperature Température d'essai	Kerbschlagarbeit Absorbed energy Energie absorbée	Kerbschlagzähigkeit Impact strength Résistance au choc	Verf.-Bruchanteil Shear fracture Rupture ductile
Anforderungen Requirements Exigences			Länge Length Longueur	Breite Width Largeur	Höhe Height Hauteur	längs (L) longitudinal (L) longitudinal (L)	einzel single individuelle	mittel average moyenne
			mm	mm	mm	quer (Q) transversal (Q) transversal (Q)	GRAD °C -45	J MIN 007
			55	5,00	10,00			J MIN 009
0020	000001	762106	55	5,00	10,00	L	-45	104
				5,00	10,00			112
				5,00	10,00			100
0030	000001	762135	55	5,00	10,00	L	-45	86
				5,00	10,00			93
				5,00	10,00			89

PO 25016962

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Kerbschlagbiegeversuch Notched bar impact test / Essai de flexion par choc (résilience) [CHARPY_V]

Pos.	Proben-Nr.	Schmelzen-Nr.	Probenabmessung			Probenlage	Prüftemperatur	Kerbschlagarbeit		Kerbschlagzähigkeit		Verf.-Bruchanteil		
Item	Specimen No.	Heat No.	Specimen dimensions			Specimen position	Test temperature	Absorbed energy		Impact strength		Shear fracture		
Poste	No. de l'éprouv.	No. de coulée	Dimensions de l'éprouvette			Position de l'éprouvette	Température d'essai	Energie absorbée		Résistance au choc		Rupture ductile		
Anforderungen			Länge	Breite	Höhe	längs (L)		einzel	mittel	einzel	mittel			
Requirements			Length	Width	Height	longitudinal (L)		single	average	single	average			
Exigences			Longueur	Largeur	Hauteur	longitudinal (L)		individuelle	moyenne	individuelle	moyenne			
						quer (Q)	GRAD							
			mm	mm	mm	transversal (Q)	°C	J	J	J/cm²	J/cm²			
			55	3,33	10,00	transversal (Q)	-45	MIN 005	MIN 006					
0010	000001	552918	55	3,33	10,00	L	-45	50						
				3,33	10,00			48						
				3,33	10,00			46	48					

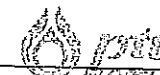
Wärmebehandlung / Heat treatment / Traitement thermique

Normalized: 900 °C

Vermerk / Remark / Remarque

NACE Standard: Hardness acc. to NACE MR0175/ISO 15156: HRC max 22.

Verkäufer(in) / Salesman/woman in charge / Personne chargée : Mrs Sehrbrock,



Chaput &

Inspection Q&R/Integrity

Integrity and Reliability

Chaput

Dinslaken, 02.02.2009,

BENTELER 
STAHLEUROPE
Benteler Stahlrohr GmbH

Abnahmebeauftragter

Inspection representative

Contrôleur

ROLF TRITTEL / WIL

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